WRITE-UP REPORT

Credit Default Risk Assessment

UCI Credit Card Default Dataset · Taiwan, 2005 · Vitto DS Internship

Author: Anuraag Juneja **Model:** XGBoost Classifier **Dataset:** 30,000 records · 25 variables

01 · APPROACH

The analysis followed a four-stage pipeline: exploratory data analysis, feature engineering, classification modelling, and communication of findings to a non-technical audience. The dataset (30,000 Taiwan credit card clients, Apr–Sep 2005) presents a binary classification problem with a class imbalance of approximately 78% non-default to 22% default.

EDA confirmed that repayment delay variables (PAY_0 through PAY_6) are the dominant signal — clients with even one month of delay show materially higher default rates. Demographic variables (SEX, MARRIAGE) showed limited predictive lift once behavioural features were included. Negative BILL_AMT values (overpayments / credit balances) were retained as economically valid observations. Undocumented EDUCATION values (0, 5, 6) and MARRIAGE value 0 were merged into an 'other' category.

02 · FEATURE ENGINEERING

Three behavioural risk indicators were engineered to compress six months of repayment history into interpretable signals:

Feature	Formula	Captures
AVG_UTIL_RATE	mean(BILL_AMTx / LIMIT_BAL)	Sustained credit pressure
AVG_PAY_RATIO	mean(PAY_AMTx / BILL_AMTx) where BILL > 0	Repayment consistency
TOTAL_DELAY_MONTHS	count(PAY_x > 0)	Cumulative delinquency

03 · MODEL DEVELOPMENT & RESULTS

A Logistic Regression baseline was trained first to establish statistical significance and a performance floor. XGBoost was then trained as the primary model, given its ability to capture non-linear interactions in repayment behaviour. Both models used **class_weight='balanced'** rather than SMOTE — synthetic oversampling within cross-validation folds risks data leakage and inflated AUC estimates; class weighting avoids this cleanly.

Stratified 5-fold cross-validation was used throughout to preserve the 22/78 class ratio in each fold.

Model	Precision	Recall	F1	AUC-ROC	CV AUC (mean ± std)
Logistic Regression	—	—	—	—	— (see notebook)
XGBoost (final)	0.5019	0.5931	0.5437	0.7807	0.7571 ± 0.0219

Top 5 features by XGBoost importance:

- **PAY_0** — Most recent repayment delay — single strongest predictor

- **TOTAL_DELAY_MONTHS** — Cumulative months with delayed repayment
- **PAY_2** — Repayment status two months prior
- **AVG_UTIL_RATE** — Average credit utilisation over 6 months
- **AVG_PAY_RATIO** — Repayment consistency relative to bill amounts

04 · TRADEOFFS & DECISIONS

class_weight vs SMOTE: SMOTE generates synthetic minority-class samples, which can leak information across CV folds if applied before splitting. class_weight='balanced' adjusts loss function weights with zero leakage risk. The modest AUC difference (~0.01) did not justify the complexity.

XGBoost vs Random Forest: Both are strong on tabular data. XGBoost's sequential boosting corrects residual errors from prior trees, yielding better recall on the minority class — important here, as missed defaults are costlier than false alarms.

Undocumented values: EDUCATION values 0, 5, 6 and MARRIAGE value 0 are absent from the official schema. Merging into 'other' is conservative but prevents the model from learning spurious patterns from artifact-level categories.

Negative BILL_AMT: Represent genuine credit balances (overpayments). Clipping to zero would destroy a signal — these clients tend to have lower default rates, which is economically meaningful.

05 · WHAT I WOULD IMPROVE WITH MORE TIME

- Calibrate predicted probabilities (Platt scaling or isotonic regression) so scores map directly to business risk thresholds — e.g. 'flag clients above 30% default probability' rather than relying on a binary cutoff.
- Run a fairness audit: compare False Positive Rates across SEX and EDUCATION groups to surface any disparate impact before deployment.
- Add SHAP explanations for individual predictions — critical for credit decisions under explainability requirements, and useful for loan officer override workflows.
- Explore temporal validation: train on Apr–Jul, test on Aug–Sep to simulate real deployment conditions rather than random cross-validation.
- Build a lightweight Streamlit app with a client input form returning default probability with a plain-language explanation of the top contributing factors.

Submitted as part of the Vitto Data Science Internship Technical Assessment. Full code, plots, and notebook available in the accompanying GitHub repository.